Krishal Hriday

Gmail | LinkedIn | GitHub

EDUCATION

PES University | B.Tech, Computer Science | CGPA: 7.47 2022 – 2026

Sri Chaitanya Techno School | Class XII 2022

Kendriya Vidyalaya IISC | Class X 2020

SKILLS

Languages: Python, JavaScript, TypeScript, C++
ML/AI: PyTorch, LangChain, CLIP/CLAP, Hugging Face Transformers, FAISS, RAG
Systems: Apache Kafka, Apache Spark, Flask, Node.js, REST APIs
Web: React.js, Next.js, Tailwind CSS
Data/Infra: PostgreSQL, MongoDB, SQLite, Git, AWS, Jupyter

ACHIEVEMENTS

- **IEEE Xplore Publication:** Authored "Brittle Semantic Collapse Under Perturbation," a study on cascade failure in multi-step LLM tool-use chains, accepted at IEEE ICWITE 2026.
- **Academic Excellence:** Awarded Distinction in the 5th and 6th semesters.

PROJECTS

Brittle Semantic Collapse Under Perturbation – IEEE ICWITE 2026

- Designed and ran a 300+ query benchmark exposing that edge-deployable 3B LLMs (Qwen2.5, Llama-3.2) collapse from 90% to 15% semantic accuracy under minor prompt perturbations despite 100% syntactic validity.
- Proved brittleness is model-intrinsic via an 81-run ablation (Welch's ANOVA $p < 10^{-83}$, $\eta^2 = 0.69$) and quantified cascade failure growth across 2–5 step tool-use chains.

Query-Conditioned Adaptive Fusion for Video Retrieval

- Built an any-to-any (text/image/audio/video) retrieval system using a learned gating network to dynamically fuse separate CLIP and CLAP embedding spaces per query.
- Designed a lightweight temporal transformer over frozen CLIP features to capture motion semantics while keeping training limited to two small trainable modules.

Distributed Image Processing Pipeline (Apache Kafka)

- Engineered a fault-tolerant, tile-parallel image processing system on Apache Kafka with manual partitioning for guaranteed 50/50 load balancing across a multi-worker consumer group.
- Built asynchronous reassembly, SQLite-backed job tracking, and a real-time Flask monitoring dashboard driven by worker heartbeats.

Fed-PEKD: Federated Pseudo-Embedding Knowledge Distillation

- Designed a data-free federated learning algorithm using per-class prototype transmission and server-side GMM sampling, cutting communication cost ∼150× versus FedAvg.
- Diagnosed and resolved server overfitting and zero-prototype crash failures through robust fallback generation, achieving stable ∼70% accuracy on non-IID FashionMNIST.